Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

Introduction to Cybersecurity

PRESENTED BY:
SEBBAR ANASS

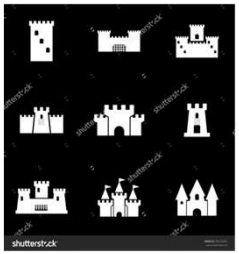
Année universitaire: 2024-2025

1

1

CyberSecurity Domains

- **Websites and Power of Data**
  - ✓ Great businesses have been created by collecting and harnessing the power of data and data analytics
  - ✓ These businesses have the responsibility to protect this data from misuse and unauthorized access
  - ✓ The growth of data has created great opportunities for cybersecurity specialists
- **Domains**
  - ✓ Business large and small have recognized the power of big data and data analytics
  - ✓ Organizations like Google, LinkedIn, Amazon provide important services and opportunity for their customers
  - ✓ The growth in data collection and analytics poses great risks to individuals and modern life if precautions are not taken to protect sensitive data from criminals or others who have intent to harm



2

2

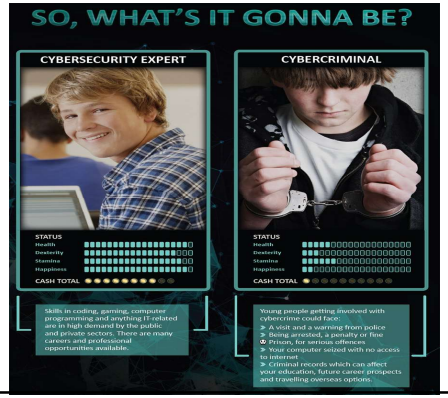
CyberSecurity Domains

- ✓ Cyber experts now have the technology to track worldwide weather trends, monitor the oceans, and track the movement and behavior of people, animals and objects in real time.
- ✓ New technologies, such as Geospatial Information Systems (GIS) and the Internet of Everything (IoE), have emerged. Each depends on collecting and analyzing tremendous amounts of data.
- ✓ This growing collection of data can help people save energy, improve efficiencies, and reduce safety risks.

3

3

Cybersecurity Criminals versus Cybersecurity Specialists



4

4

Cybersecurity Criminals versus Cybersecurity Specialists

5

5

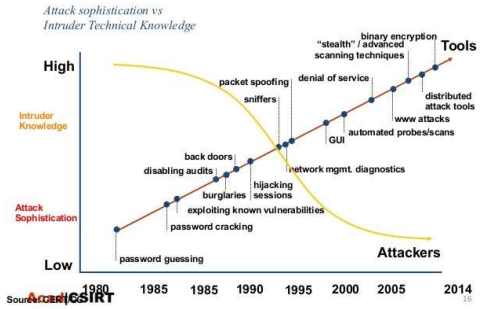
Cost of IT Security per Year

Type de crime informatique	Royaume Unis	Monde
Online banking fraud		
Phishing	$16m	$320m
Malware (consumer)	$4m	$70m
Malware (business)	$6m	$300m
Bank tech. countermeasures	$50m	$1000m
Cost of cyber criminal infrastructure		
Expenditures on antivirus	$170m	$3400m
Cost to users of clean up	$500m	$10000m
Defense costs of firms	$500m	$10000m

Estimation de certains coûts associés à la Security informatique.
Source: Anderson et al. Measuring the cost of cybercrime. WEIS 2012:

6

No need to be an Expert to Launch Attacks ☺



Security tools allow hackers to launch attacks with little technical

7

Types of hackers

- Script kiddies
  - beginner hackers
  - Use ready-to-use software
- Crime informatique organized
  - Think the mafia. Ex: Russian Business Network
  - Monetary purpose
  - Very sophisticated and specialized: Virus development, virus distribution, hosting, etc.
- States
  - Gouvernments e.g. USA, Russie, Chine, Ukraine etc
  - But politique: espionage, sabotage, etc
  - Sophistication sans précédant

8

Types de Pirates

- Hackactivists
  - IT activist: political goals
  - Example: Anonymous
  - Sophistication varies
- White hats
  - Ethical hacker
  - Goal: discover vulnerabilities before malicious hackers

9

CyberSecurity

10

10

CyberSecurity Terminologies

Common network security terms:

- **Attack**: Any action that compromises the security of information.
- **Security mechanisms:** a mechanism that is designed to detect, prevent and combat a security attack.
- **Security Service:** a service that increases the security of data processing and exchange of a system. A security service uses one or more security mechanisms.

11

11

CyberSecurity

The Steps of an attack?

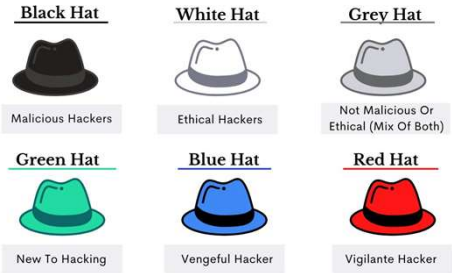
- Example of a real attack Phishing vs Ransomware

12

12

Types de Pirates

- **Hackers** – This group of criminals breaks into computers or networks to gain access for various reasons.
  - ***White hat*** attackers break into networks or computer systems to discover weaknesses in order to improve the security of these systems.
  - ***Gray hat*** attackers are somewhere between white and black hat attackers. The gray hat attackers may find a vulnerability and report it to the owners of the system if that action coincides with their agenda.
  - ***Black hat*** attackers are unethical criminals who violate computer and network security for personal gain, or for malicious reasons, such as attacking networks.



13

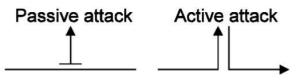
13

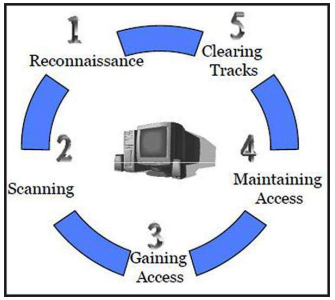
CyberSecurity

The Steps of an attack

1. Reconnaissance
2. Scanne (adresses, port, vulnérabilités)
3. Gagner access
4. Maintaing access
5. Clearing Tracks

Passive Attacks and Active Attacks





14

14

Passive and Active Attacks

Passive Attack	Active Attack

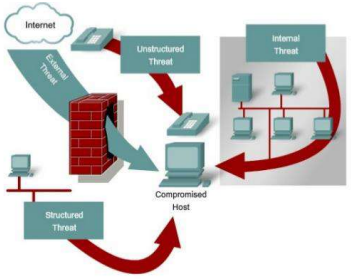
• Attempts to learn or make use of information from the system but does not affect system resources
• Eavesdropping on, or monitoring of, transmissions
• Goal of attacker is to obtain information that is being transmitted
• Two types:
  • Release of message contents
  • Traffic analysis
	• Attempts to alter system resources or affect their operation
	• Involve some modification of the data stream or the creation of a false stream
	• Four categories:
	  • Replay
	  • Masquerade
	  • Modification of messages
	  • Denial of service

15

CyberSecurity

Vecteurs d'attaques de réseau

- Attaques externes
- Attaques internes
- Attaques structurées
- Attaques non structurées



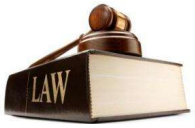
16

16

The laws of CyberSecurity

Legislation:
Several laws are established by the federal and government to protect users from the threats of cybercrime:
1- The healt Insurance Portability and Accountability 1996
2- The sarbanes-Oxley Act of 2002
3- The Gramm-Leach-Blilely Act
4- US PATRIOT ACT 2001

In Morocco we have:
1- Law 53-05 on the electronic exchange of legal data
2- Law 09-08 on the automated processing of personal data

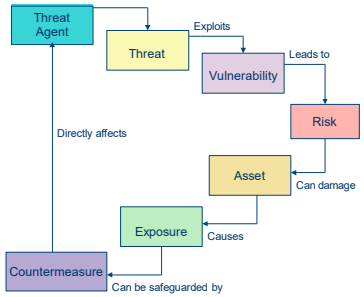


17

17

CyberSecurity

Common network security terms:



18

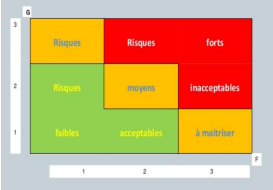
18

CyberSecurity Risk

Risk Management
Definition:
The assessment and quantification process and the establishment of an acceptable level of risk to the organization

Risk = Probability * Impact

Stages of risk management:
- **Risk analysis**
- **Assessed and identified threats**
- **Assessed and identified vulnerabilities**
- **Establish a Security Policy (countermeasure)**



19

19

CyberSecurity Risk

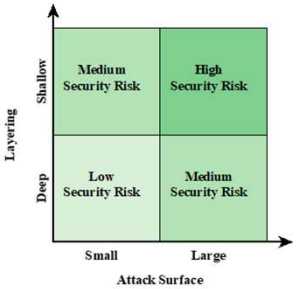


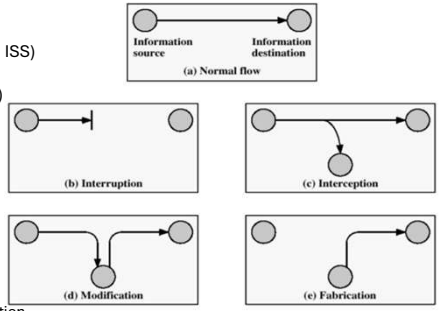
Figure 1.4 Defense in Depth and Attack Surface

20

CyberSecurity

Attackers' tools

- Port/address scan (nmap, Nessus)
- Vulnerability Scanner (MetaSploit, Core Impact, ISS)
- Sniffers (Wire shark, tcpdump, Snort)
- Cracking tools (Cain, wepCrack, jhon the ripper)
- Malware (viruses, worms, trojon)
- Hijackin tools (Netcat, MetaSploit)



Aims of attacks

- **Interruption**: aims at the availability of information
- **Interception**: aims at the confidentiality of information
- **Modification**: aims at the integrity of information
- **Manufacturing**: aims at the authenticity of the information